

SC Honors Geometry - Nappier
Homework 10.6 – Segment Relationships in Circles

Name: _____

Two chords: (piece)(piece) = (piece)(piece).

1. Pieces 5 and 8; pieces x and 4. Find x .
2. Pieces 7 and 6; pieces 14 and x . Find x .

Two secants: (whole)(external) = (whole)(external).

3. External 5, far part 11; external 4, far part x . Find x .
4. External 6, far part 8; external 7, far part x . Find x .

Tangent and secant: $\text{tangent}^2 = (\text{whole})(\text{external})$.

5. External 9, far part 16. Find the tangent t .
6. Tangent 12; external 8, far part x . Find x .

Mixed practice.

7. Chord pieces 3 and $(x + 4)$; other chord pieces 6 and 5. Find x .
8. Tangent 20; secant external x , far part 30. Find x .
9. A broken circular gear: two half-chords of 12 meet a diameter piece of 4 on one side and x on the other. Find x and the diameter.
10. Two secants from P : external 8, far part 7; external 10, far part x . Find x .